

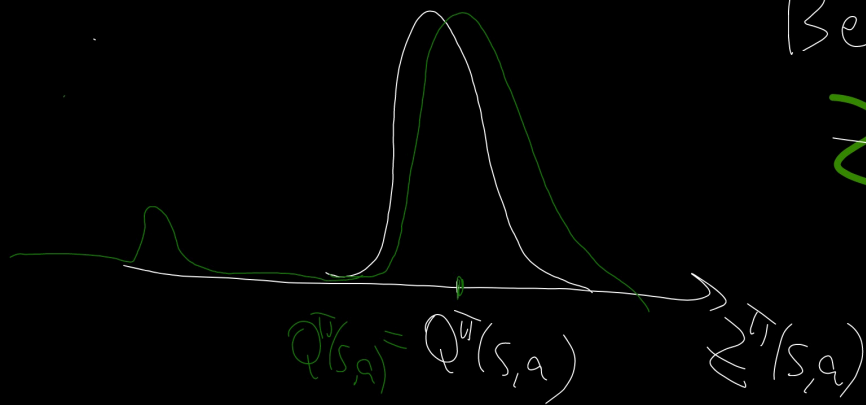
Distributional RL

- Risk-sensitive agents

$$Z^\pi(s, a) = r(s, a) + \gamma Z(s', a') + \gamma^2 Z(s'', a'') + \dots$$

$$Q^\pi(s, a) = \mathbb{E} Z^\pi(s, a)$$

$$s' \sim p(s' | s, a) \\ a' \sim \pi(a' | s')$$



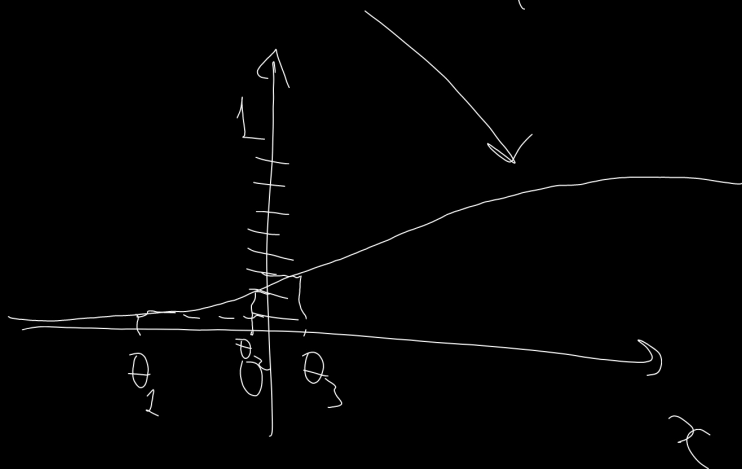
Bellman equation:

$$Z^\pi(s, a) = r(s, a) + \gamma Z^\pi(s', a')$$

= by dist. $s' \sim p(s' | s, a)$
 $a' \sim \pi(a' | s')$

QR - DQN

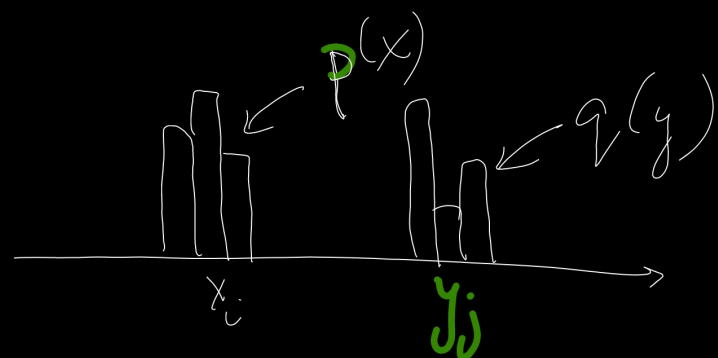
$$\text{cdf } F(x) = \mathbb{P}\{X < x\}$$



$$p_X(x) \approx \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i}(x)$$

$$KL(p||q) = \int p(x) \log \frac{p(x)}{q(x)} dx$$

Wasserstein distance



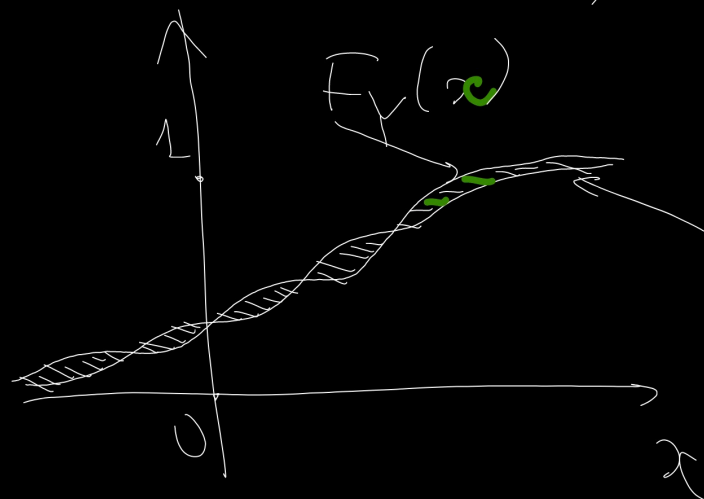
$\gamma(x_i, y_j)$ — transportation plan

$$W(p, q) = \inf_{\gamma} \sum_{i,j} \gamma(x_i, y_j) \|x_i - y_j\|$$

$$\begin{aligned} \gamma(x_i, y_j) &\geq 0 \quad \forall i, j \\ \sum_i \gamma(x_i, y_j) &= q(y_j) \quad \forall j \\ \sum_j \gamma(x_i, y_j) &= p(x_i) \quad \forall i \end{aligned}$$

cdf $F_X(x) = P\{X < x\}$, $F^{-1}(\tau) = \inf_x \{F(x) \geq \tau\}$

Th $W(X, Y) = \int_0^1 |F_X^{-1}(\tau) - F_Y^{-1}(\tau)| d\tau$

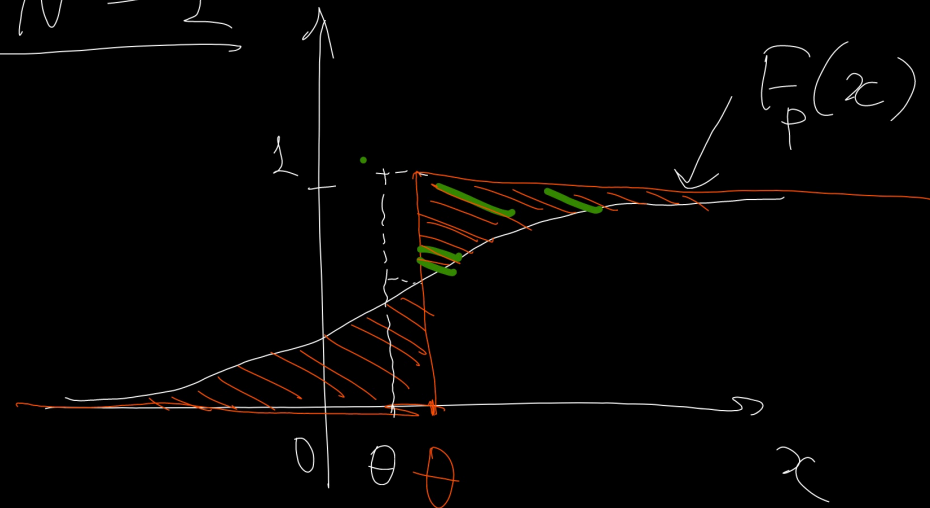


$F_X(x)$

$W(X, Y) = \text{Area}$

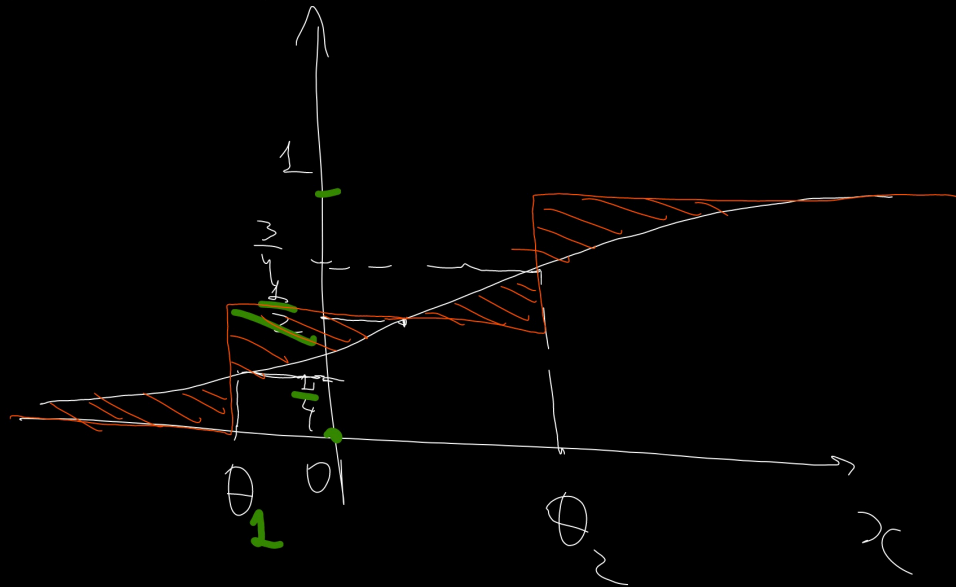
$$p(x), \quad q(x) = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i}(x), \quad W(p, q) \rightarrow \min_{\{\theta_i\}}$$

$$N=1$$



$$\theta = \text{median}(p)$$

$$\underline{N = 2}$$



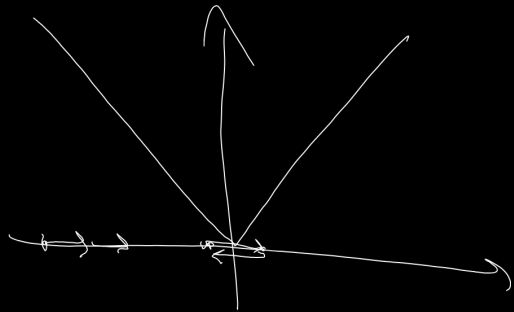
$$N \text{ opt } \theta_i =$$
$$= \hat{c}_i - \text{quantile}$$
$$\hat{c}_i = \frac{2i-1}{2N} \quad i=1, 2, \dots, N$$

Quantile estimation

$$x_1, x_2, \dots, x_N \in \mathbb{R}$$

$$\theta = \frac{1}{N} \sum_{i=1}^N x_i \Rightarrow L(\theta) = \frac{1}{N} \sum_{i=1}^N (\theta - x_i)^2 \rightarrow \min_{\theta}$$

$$\theta = \text{median}(x_1, \dots, x_N) \Rightarrow L(\theta) = \frac{1}{N} \sum_{i=1}^N |\theta - x_i| \rightarrow \min_{\theta}$$



Huber loss $\mathcal{L}_\varepsilon(u) = \begin{cases} \frac{u^2}{2}, & |u| \leq \varepsilon \\ \varepsilon|u| - \frac{\varepsilon^2}{2}, & \text{otherwise} \end{cases}$

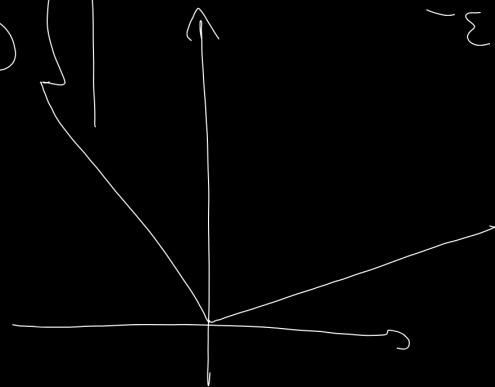
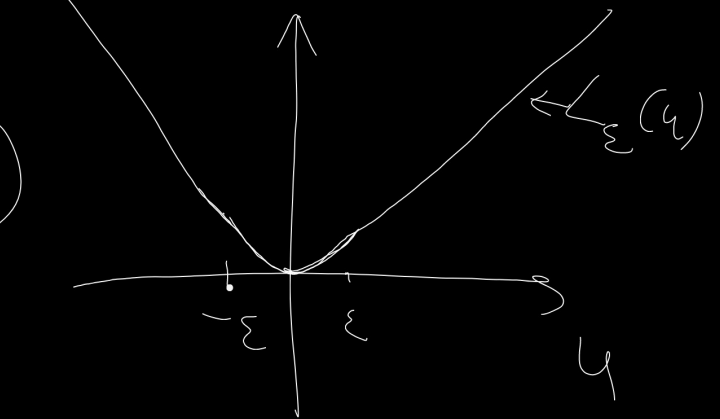
$\frac{1}{N} \sum_{i=1}^N \mathcal{L}_\varepsilon(\theta - x_i) \rightarrow \min_{\theta}$

check loss $\Rightarrow \theta_{\text{opt}} \approx \text{median}(x_1, \dots, x_N)$

$\mathcal{L}_\tau(u) = |u| \left| \tau - [u < 0] \right|$

$\tau = \frac{1}{4}$

$u \geq 0$	$u \cdot \frac{1}{4}$
$u < 0$	$-u \cdot \frac{3}{4}$



$$L_{\tau, \varepsilon}(u) = L_{\varepsilon}(u) \cdot |\tau - [u < 0]|$$

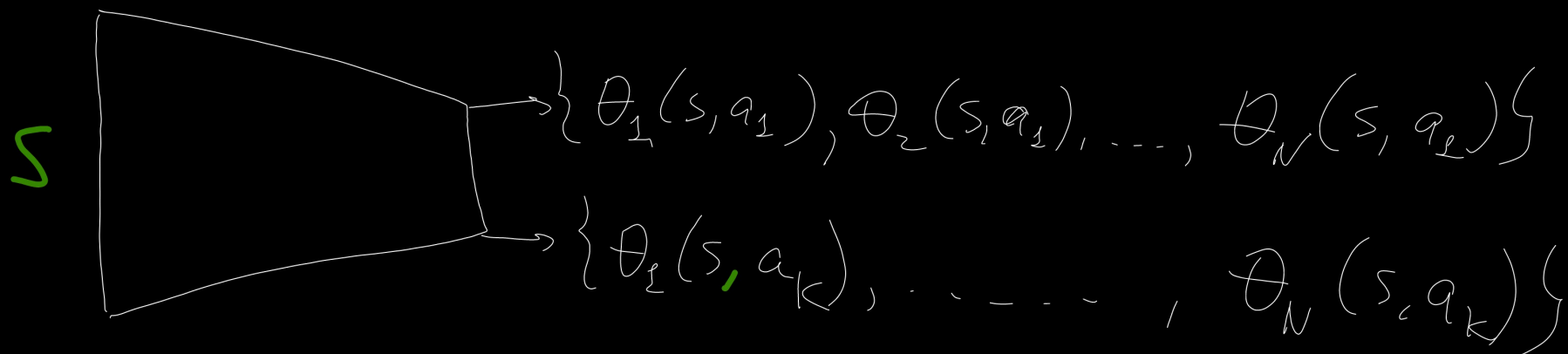
$$\frac{1}{N} \sum_{i=1}^N L_{\tilde{\tau}, \varepsilon}(\theta - x_i) \rightarrow \min_{\theta} \Rightarrow$$

•

$$\text{In practice } \varepsilon = 1 \Rightarrow \theta_{\text{opt}} \sim \tilde{\tau}\text{-quantile}(x_1, \dots, x_n)$$

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QR-DQN



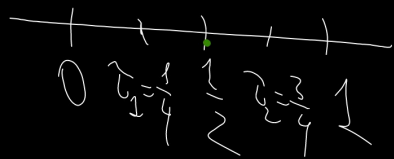
$$\sim s, a, z, s' \quad a' = \arg \max_{a'} \frac{1}{N} \sum_{i=1}^N \theta_i(s', a' | \omega)$$

$$y_j(s, a) = z + \gamma \theta_j(s', a' | \omega_{\text{target}}), \quad \forall j = 1, \dots, N$$

$$L(\omega) = \frac{1}{N} \cdot \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N L_{\tau_i, \varepsilon} \left(y_j(s, a) - \theta_j(s, a | \omega) \right)$$

Test: $\arg \max_a \sum_i \left(\psi(b(\tau_i)) - \psi(a(\tau_i)) \right) \times \theta_i(s, a)$ $\tau_i = \frac{z_i - 1}{2N}, i = 1, \dots, N \rightarrow \text{high } \omega$

$b(\tau_i) = a(\tau_{i+1})$

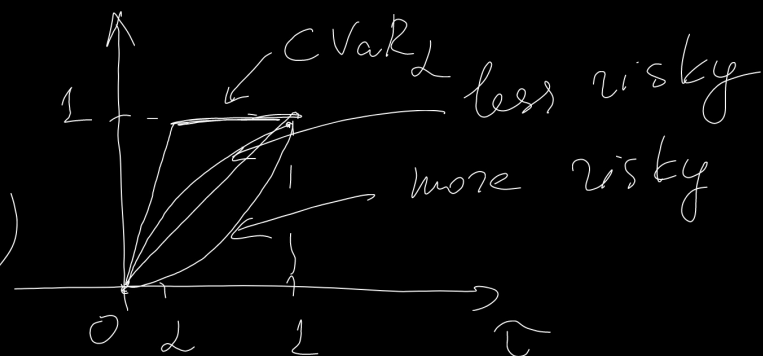


Testing

Wand risk:

$$\psi_1(\tau) = \Phi(\Phi^{-1}(\tau) - d)$$

cdf for $N(0,1)$



$$a = \arg \max_a CVaR_L(\{\theta_i(s, a)\})$$

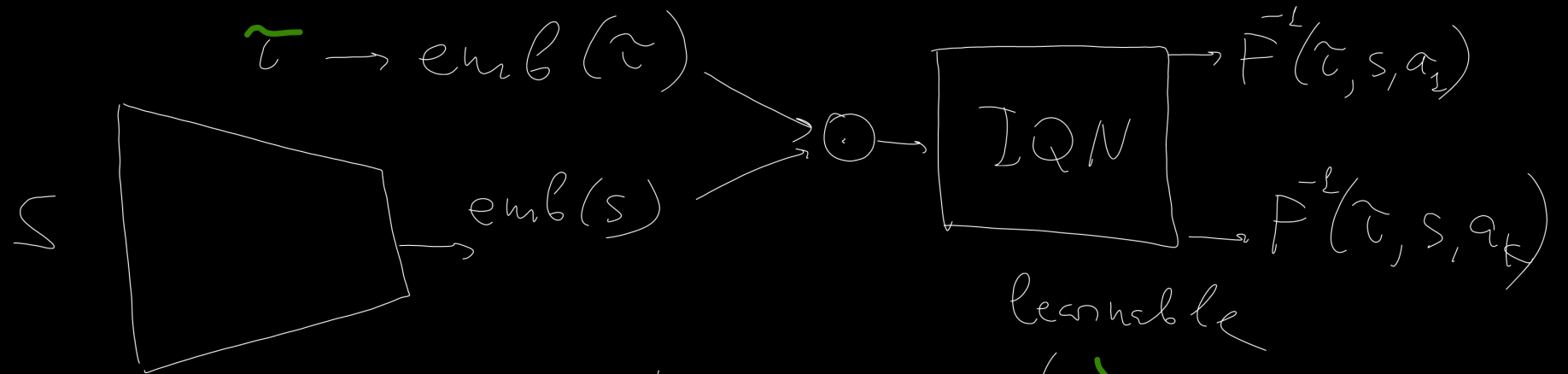
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take first $[LN]$ quantiles

distortion risk measure $\psi(\tau): [0,1] \rightarrow [0,1]$ and average

$$a = \arg \max_a \int_0^1 F^{-1}(\tau, s, a) \psi'(\tau) d\tau$$

IQN



$$emb_j(\tau) = \text{ReLU}\left(\sum_{i=1}^n \cos(\pi i \tau) w_{ij} + b_j\right), \quad n=1 \dots 64, \quad j=1, \dots, \text{size}(emb(s))$$

$$\sim s, a, r, s'$$

$$a' = \arg \max_{a'} \frac{1}{K} \sum_{k=1}^K F^{-1}(\hat{\tau}_k, s', a')$$

$$\hat{\tau}_k \sim U[0, 1]$$

$$K=32$$

$$\tilde{\tau}_j \sim U[0, 1], j=1, \tilde{N}$$

$$\tau_i \sim U[0, 1], i=1, N$$

$$Loss = \frac{1}{N} \frac{1}{\tilde{N}} \sum_{i=1}^N \sum_{j=1}^{\tilde{N}}$$

$$N = \tilde{N} = 8$$

$$L_{\tau_i, \varepsilon} \left(r + \gamma F^{-1}(\tilde{\tau}_{ij}, s', a' | \sigma_{\text{target}}) - F^{-1}(\tau_i, s, a | \sigma) \right) \rightarrow \min_{\sigma}$$